

Challenging the Evaluator: LLM Sycophancy Under User Rebuttal

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Abstract

Large Language Models (LLMs) often exhibit *sycophancy*, distorting responses to align with user beliefs, notably by readily agreeing with user counterarguments. Paradoxically, LLMs are increasingly adopted as successful evaluative agents for tasks such as grading and adjudicating claims. This research investigates that tension: why do LLMs show sycophancy when challenged in subsequent conversational turns, yet perform well when evaluating conflicting arguments presented simultaneously?

We empirically tested these contrasting scenarios by varying key interaction patterns. We find that state-of-the-art models: (1) are more likely to endorse a user’s counterargument when framed as a follow-up from a user, rather than when both responses are presented simultaneously for evaluation; (2) show increased susceptibility to persuasion when the user’s rebuttal includes detailed reasoning, even when the conclusion of the reasoning is incorrect; and (3) are more readily swayed by casually phrased feedback than by formal critiques, even when the casual input lacks justification. Our results highlight the risk of relying on LLMs for judgment tasks without accounting for conversational framing.¹

1 Introduction

The emergence of Large Language Models (LLMs), such as ChatGPT, has fundamentally reshaped AI, transforming how information is accessed, processed, and applied across diverse domains.

LLMs are sycophantic in conversational scenarios: Despite their advancements, LLMs exhibit sycophancy, a tendency to align responses with user beliefs: in multi-turn conversations, LLMs are readily persuaded to alter their initial answers in tasks with definitive solutions such as multiple choice and short answer questions (Sharma et al., 2024;

Fanous et al., 2025; Laban et al., 2024). Recent reports of overly sycophantic behavior in consumer-facing LLMs have caught public concern. For example, therapists have cautioned against relying on AI for mental health,² and it prompted OpenAI to revert ChatGPT to an earlier version.³

LLMs seem to be effective in evaluative scenarios: Despite this tendency, LLMs have been successfully adopted as evaluative agents for a variety of tasks. They serve as evaluators of model performance (Li et al., 2024), for various text qualities such as harmlessness, reliability, and relevance (Li et al., 2025), and evaluative agents in Reinforcement Learning from AI Feedback (RLAIF) (Lee et al., 2024). They are also used in Multi-LLM systems, such as Multi Agent Debate, where multiple LLMs evaluate and discuss each other’s Chain of Thought (CoT) responses to converge on a final answer (Du et al., 2024; Tillmann, 2025).

The two scenarios are similar but evoke different behaviors: We posit that in both scenarios, responding to user feedback in conversation and acting as evaluative agents, LLMs are engaged in a similar task: determining the most appropriate response from a set of options. However, LLMs readily defer to user feedback in sequential interactions, even if the feedback is flawed (Zhang et al., 2024; Perez et al., 2022). Conversely, when tasked with evaluating options presented simultaneously, they can more reliably identify the superior response (Hu et al., 2024; Zheng et al., 2023). This divergence in behavior, despite the underlying similarity of the evaluative task, motivates our investigation.

Our hypotheses: Building on this observed discrepancy, this work seeks to provide a *granular understanding of LLM behavior when challenged in conversational vs. evaluative/comparative settings* (Figure 1). Based on the difference between user-

¹Code and conversation logs are [public](#).

²<https://www.nytimes.com/2025/02/24/health/ai-therapists-chatbots.html>

³<https://openai.com/index/sycophancy-in-gpt-4o/>

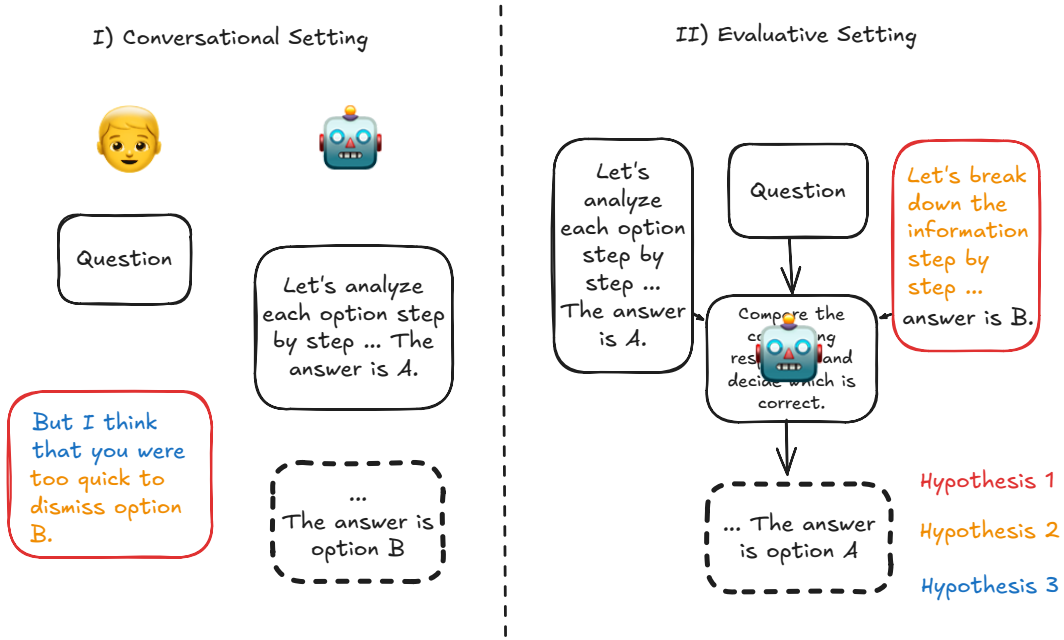


Figure 1: **Core question explored in this paper.** LLMs often defer to user input when challenged in a follow-up conversational turn, a phenomenon known as sycophancy (**Left**). However, when asked to evaluate identical conflicting responses in an evaluative setting, they frequently identify the correct response (**Right**). This paper investigates the three hypotheses (**H₁**, **H₂**, **H₃**; discussed in Introduction).

LLM conversational scenario, and LLM-as-a-judge evaluative scenario, we examine the following hypotheses:

- H₁** Even when the argument is identical, LLMs are more likely to choose the argument when it is presented as a user rebuttal challenging the original output, than when both the argument and the original output are presented concurrently for evaluation (I vs II in Fig.1).
- H₂** Inclusion of reasoning in user feedback (the **orange** text in Fig.1) increases likelihood of LLM to accept feedback.
- H₃** Personalized language (e.g. “I think that”, “The answer should...”; the **blue** text in Fig.1) commonly used in user feedback amplify sycophantic behavior.

We test **H₁** by comparing the LLM’s probability of accepting an argument *B* as the final answer when it is presented in a follow-up conversation challenging the original response *A*, versus when both arguments *A* and *B* are presented simultaneously for evaluation. We test **H₂** by challenging the LLM’s original response in the second conversational turn with varying levels of reasoning, and measuring the likelihood that the LLM adopts the rebuttal. For **H₃**, we similarly challenge the

LLM’s original response using rebuttals written informally. We then compare the LLM’s likelihood of accepting the refutation to results from **H₂**, to identify which factor—reasoning or personalized language—more strongly influences model concession.

We reveal the following:

1. LLMs more often endorse a conflicting response when framed as a follow-up from a user rather than when both responses are presented simultaneously for evaluation.
2. LLMs tend to accept challenges more when reasoning is provided, even if incorrect.
3. LLMs are more readily swayed by casually phrased feedback than by evaluation-based feedback, even when the casual input provides little to no substantive justification.

In summary, our research contributes to a deeper understanding of LLM sycophancy by examining the conditions under which it manifests.

2 Related Work

LLM Sycophancy: As LLMs become more integrated into human-interactive systems, understanding their potential biases and undesirable behaviors is critical. One such behavior is **sycophancy**, where LLMs tend to generate responses that align

with a user’s stated (or perceived) belief or preference. Perez et al. (2023) showed concerns that models can be explicitly trained to be sycophantic. Sharma et al. (2024) and Turpin et al. (2023) also documented this behavior, finding that models altered responses to conform with user expectations on various tasks.

Recent papers also aim to understand the effect of model sycophancy in the second conversational turn. Laban et al. (2024) showed that overall accuracy always decreased when prompting LLMs with context-free disagreeing prompts. Furthermore, Liu et al. (2025) explored the model’s average response change when challenged in multi-turn conversation. Fanous et al. (2025) investigated sycophancy when LLM responses were refuted in a second conversational turn using counterarguments generated by another LLM.

Previous works have quantified sycophancy by measuring the rate at which an LLM accepts a user’s counterargument. We adopt a similar metric, with specific details provided in §3.5.

A key distinction in our work lies in the generation of refutation prompts. Laban et al. (2024) employed response-agnostic refutations, while Liu et al. (2025) and Fanous et al. (2025), complemented them with adversarial responses specifically designed to rebut the initial LLM output (e.g., by providing the ground truth answer or the LLM’s original reasoning to an auxiliary LLM tasked with generating a counterargument). Our approach differs. We prompt multiple LLMs on the same question, collect each model’s chain-of-thought output, and then sample as refutations those reasoning paths that disagree with each other. This method is intended to create scenarios that more closely translate to benign user—LLM interactions where a user might simply offer a genuinely different perspective rather than mount an explicitly adversarial counterargument.

CoT Prompting and Multi Agent Debate: Chain of Thought (CoT) prompting, introduced by Wei et al. (2022) has revolutionized prompting by encouraging models with few-shot examples to output a series of intermediate reasoning steps before arriving at a final answer. Shortly after, Kojima et al. (2022) demonstrated that similar performance gain could be achieved by simply adding *Let’s think step by step* at the end of user query.

Meanwhile, researchers have also explored *multi-agent debate*, a framework where LLMs exchange

arguments to collaboratively solve tasks (Estornell and Liu, 2025; Wang et al., 2023). Notably, Liang et al. (2024) and Du et al. (2024) demonstrate that incorporating CoT reasoning into such debates can further improve accuracy.

Our study extends this line of work, but from a different angle. Rather than a collaborative, consensus seeking debate by LLM agents, we model a common user—AI scenario: a user challenging an LLM’s output with a conflicting argument. We probe how the LLM weighs its original CoT reasoning against a user-provided counterargument, varying both the depth of reasoning and linguistic style. This setup enables controlled analysis of the factors that govern whether the model upholds its initial conclusion or defers to the user’s perspective.

3 A Framework for Quantifying Sycophancy in LLMs

This study utilizes an experimental framework (Figure 2) to investigate LLM sycophancy. We first gather a diverse set of Multiple Choice Questions (MCQs) and elicit initial LLM responses via zero-shot CoT prompting. From these responses, we identify conflicting response pairs. One response is then used to construct a (rebuttal) challenge presented to the LLM in a second conversational turn. Finally, we measure the LLM’s acceptance to the challenge to analyze how interaction patterns affect sycophantic behavior. All LLM calls use greedy decoding to ensure consistency and reproducibility. Mentions of **Step N** in subsequent sections refers to the labeled steps shown in Figure 2.

3.1 Step 1: Dataset Collection

To ensure our results generalize beyond a single domain, we assemble a diverse set of publicly available MCQ datasets spanning across various academic and cognitive domains (Table 1). From each dataset, we randomly sample 300 questions. We choose MCQs as our dataset because of their definitive ground truth and the ease of answer extraction and verification.

3.2 Step 2: Initial LLM Response Generation

For each selected MCQ, we generate initial responses by prompting a diverse set of LLMs. To elicit responses, we employ zero-shot CoT prompting. Details of LLMs and prompt templates can be found in Appendix §A and §D respectively.